

Lab 1: Bash Practice

Instructions

Log in to the classes.csc server through PuTTY or another SSH client

Programs you will interact with for this lab are found in my lab folder.

The path to this is /classes/cs3304/cs3304_don/class/lab_1/

Or from your home directory: ~/../cs3304_don/class/lab_1/

NOTE: YOU WILL LIKELY NEED TO LOOK AT MAN PAGES / GOOGLE TO ASSIST.
THIS IS NORMAL!

Part 2: Lea

You made it into the club for the cool kids!

One problem though: When you finally got in the door, you left it open.

Now the cats from the club are running rampant on the server!

Impeccable work, I could not do it better myself!

After talking to the owner, there are 4 cats missing:

- (1) Lea: the LEAders of the cats. The other cats will follow wherever Lea goes
- (2) Clea: Clea is a bit of a CLEptomaniac! She stole one of my files and took it with her!
- (3) Ditzea: the clumsy cat of the group. Wherever he goes, things usually break!
- (4) Onomonopia: "Onny" for short, is an incredibly vocal cat. Rumor has it you can hear his meows even after he's left!

Now its YOUR responsibility to **find** the cats.

The cats have taken up residence somewhere on the classes server. They are sure to have created many of their own files to set up their homes.

Thankfully you can use the **find** command to hunt for files on a Linux system (whew). The syntax for **find** can be a bit complicated from looking at the man page. Generally, the format is `find <search location> <search parameters>`. The search parameters work like options

for other commands. One such example is `$ find ~/ -name "cat"`, which searches your home directory for any files named "cat". But searching for an exact match on a file name isn't that useful! And in come wildcards:

Token	Purpose	Example	Explained
*	Any length or content wildcard	Cat.*	Files named "Cat" with any extension
?	Single character wildcard	Letterhead-??	Any file named "Letterhead-" with exactly two characters following the dash
[chars]	Range of values	[abc]	Matches anything containing an a, b, or c
[!chs]	Anything not in range	[!abc]	Matches anything NOT a, b, or c
[[:class:]]	Matches anything in class		
Classes			
[[:alnum:]]	Any alphanumeric		
[[:digit:]]	Any digit (0,1,2...)		
[[:lower:]]	Any lowercase letter		

These wildcards are expanded by your shell into the command you input. So if you want to list all of the text files in a directory, you could type `ls *.txt`

Question 1: Using the `find` command, figure out where Lea took the cats on the server! (Hint, they're a bit egotistical, so chances are their name is in the file `@~`). Under what directory did Lea take the cats?

/tmp/Lea_Cat_Club

Example find usage: (You will need to tweak the parameters...)

```
$ find ~/ -name "cat"
```

Once you've found their hiding place, take a look around the environment! You must use your **cat** senses to learn about the various objects (files).

Question 2: Find each cat's room. List the path to their room starting from the Living Room (LivingRoom/...)

Lea: LivingRoom/LeaRoom

Clea: LivingRoom/Kitchen/CleaRoom

Ditzea: LivingRoom/Upstairs/DitzeaRoom

Onny: LivingRoom/Upstairs/OnnyRoom

Part 3: Ditzea

Ditzea is a bit, well DITtsy! He has broken so many thing inside of this poor home!

Question 3: How many different things (files) did Ditzea break?

11

Question 4: Write a command or command series to locate all of the objects broken by Ditzea, give their location, and provide a glance at the number of objects he broke. (hint: think about numbering the lines of output)

Find . -name "*.brkn"

Find . -name "*.brkn" | wc -l

Part 4: Onomonopia

Onny is a very loud boi. He meows so much, you can hear echos of his meows even after he's gone!

Question 5: Write a command to find all of the places Onny has meowed.

Find . -name "meow.echo"

Question 6: It seems that the more Onny spends in a place, the more meows are in an echo! Figure out how to measure the number of meows from a given echo.

hint: `find <location> <search> -exec <command to run on all found files> '{}'` \;

```
find . -name "*echo" -exec wc -l '{}'
```

 \;

Question 7: Find a way to chain together these two commands so that we can see all the places Onny has been, and how much time he spends in each place! Finally for good measure, we should **sort** this output so we can see all his favorite places!

```
find . -name "*echo" -exec wc -l '{}'
```

 \; | `sort -n -r`

Part 5: Clea

I think Clea has stolen one of my files! She has a box in her room that she keeps all the things she steals. However, she is a bit moody so she hisses at me if I try to approach the box. Maybe you have better luck with cats than I do?

Question 8: Go to Clea's Room, you can interact with her by running `Clea.sh`. What happens when you "interact" with Clea?

She started swinging hands

Question 9: So your luck may not be better than mine, but thankfully Clea is very food driven! She will happily eat a treat file, so see if making one helps her mood. (treat files are named "treat.X" where X is the flavor of the treat). Do you have better luck if you feed her a chicken treat?

Still throwing hands

She ate the treat! But she's still mad at you! Maybe a different flavor would work better on her.

Question 10: Find the treat flavor that Clea likes and feed it to her. She is known to audibly purr when given her favorite treat. What flavor does she like?

Fish

You made friends with Clea, and she happily placed the file she stole in your home directory!

Return to your home directory (cd ~ is quick!)

Question 11: What type of file did she steal?

PNG

Question 12: Transfer the file to a machine where you can view this file type. Did she steal this file from me? If not, who is this originally from?

Toasty

Bonus (+2 for each CORRECT answer, Maximum +10)

B1: Who is alleged treat thief? Who is the actual treat thief?

Onny alleged thief, Clea and Ditzea actual thieves

B2: What were you supposed to touch? What happened after you touched it?

Could not touch grass

B3: What is the brand of Meow-liner used?

Maypurrlane

B4: What Beloved Disney movie was alluded to?

Lion King

B5: What warning was posted about Onny and Ditzea?

They play League of Legends